Damian **Sowinski**, Ph.D.

Physicist & Applied Mathematician | Complex Systems, Computation, Simulation, & Artificial Life

Contact
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Webpages
DamianSowinski.com
OwlsNestCreations.com

Languages
English *native*
Polish *fluent*

Programming
MATLAB, Python
Mathematica,
Fortran
C, C++

Tools
GitHub
HPC:
Discovery (Dartmouth)
BlueHive (Rochester)

External Links
arXiv
ORCID
Lectures
Wordpress
Visualizations
Google Scholar

Code Repos
pyCE
pyCA
pyGD
pyCoop
pyEDW
LEAFS
AHLCG

Summary

A theoretical physicist by training and a generalist by conviction — I study how information and computation animate matter, from foraging agents and biospheres to phase transitions and spacetime. My mantra is that *everything is interesting*, and for a decade I have followed it across complex systems, statistical physics, information theory, and artificial life, building open computational tools and publishing in *The Astronomical Journal*, *The Planetary Science Journal*, *PRX Life*, *Physical Review D*, *and Nature Communications Physics*. Alongside the research, I bring fifteen years of teaching across every level, from high-school physics and mathematics to undergrad courses and labs, as well as my conviction that the best way to learn STEM is to do it. Check out my physics inspired game studio.

Education

Doctor of Philosophy**,** ***Physics*** — Dartmouth College, New Hampshire
Bachelor of the Arts**,** ***Physics*** — University of California, Berkeley

Research

Department of Physics and Astronomy, University of Rochester — Rochester, New York **2023-Present**

Research Scientist

- Co-author a comprehensive review of the history, science, and speculation on and around technosignatures with Clement Vidal's PSETI collaboration.
- Implemented a stochastic Daisyworld model and information theoretic analysis toolkit for application to exoplanets around M-dwarf stars. Collaborator: Adam Frank
- Built the LEAFS suite, an agent-based ecological simulator, with applications towards collective learning. Analyzed large GPS datasets of homing pigeon flights with Takao Sasaki and Dora Biro.
- Created an agent-based variant of the Kuramoto model, derived its mean field behavior, and performed comprehensive information theoretic analysis. Collaborator: Gourab Ghoshal
- Developed a quantum sampling algorithm for Haar random matrices with Alice Quillen.

Thayer School of Engineering, Dartmouth College — Hanover, New Hampshire **2019-2022**

Research Scientist I-III

- Designed and built a Raspberry Pi controlled sensor setup for measuring the hydraulic permeability of various porous materials, used for calibrating poroelastic MRE inversion.
- Fabricated 3D printed mechanical setups for deforming materials used in MRE experiments on the acoustoelastic effect.
- Developed a parallel finite element suite for forward simulation and inverse ML inference of poroelastic and anisotropic Neo-Hookean continuum models, with applications to tissue property imaging. Collaborators: Matthew McGarry, Keith Paulsen
- Built a computational suite to simulate and analyze magnetic nanoparticles in rapidly varying magnetic fields for application to tissue heating (hyperthermia). Collaborator: John Weaver
- Developed an ML method for image assessment which combined biological architecture with the analytical tools of differential geometry. Applied to human image similarity judgements in psychophysics experiments. Collaborators: Richard Granger, Elijah Bowen

Research (Continued...)

Institute for Cross-Disciplinary Engagement, Dartmouth College Hanover, New Hampshire **2017-2019**

Postdoctoral Scholar

- Created the mathematical foundation for Configurational Information Measures (CIM).
- Built the pyCE python library to analyze CIMs in the Landau-Ginzburg system, boson stars, instantons, oscillons, and the CMB. Advisor: Marcelo Gleiser
- Analyzed NBA video data for applications to social dynamics. Collaborator: Seth Frey

WisPIC, University of Wisconsin Madison, Wisconsin **2016-2017**

Postdoctoral Researcher

- Analyze information measures applied to Ising, Potts, and Heisenberg models on lattices and Erdős–Rényi graphs representing distinct neurotopologies; maintain a statistical mechanics pipeline for data generation and analysis.
- Collaborate with S. Streipert, applying Integrated Information Theory to split-brain phenomenon using deterministic and stochastic networks; work with L. Barbossa on spin glass systems for complex network dynamics.

Beyond Institute, University of Arizona Tempe, Arizona **2015**

Visiting Scholar

- Collaborate with S. I. Walker, H. Kim, & A. Adams on information processes in biological systems; develop a stochastic cellular automata model combining deterministic and Ising-like dynamics, analyze competition between deterministic and stochastic subprocesses, and apply machine learning to the yeast cell growth cycle

Department of Physics & Astronomy, Dartmouth College Hanover, New Hampshire **2011-2016**

Graduate Student

- Develop Configurational Entropy in statistical field theory with M. Gleiser; gain expertise in computational physics, numerical methods, and optimization across multiple programming languages and simulated systems

Hayden Planetarium, American Museum of Natural History New York, New York **2010-2011**

Research Assistant

- Work with M. Mac-Low & C. McNally on modified GADGET code in hydrodynamic simulations

Department of Astronomy, Hunter College New York, New York **2010-2011**

Research Assistant

- Analyze M and L dwarf spectra, characterizing populations in color space for future LSST photometric surveys

Lawrence Berkeley National Laboratory Berkeley, California **2000-2003**

Undergraduate Researcher

- Calibration of KamLAND photomultiplier tubes and z-axis arm under H. Steiner

Teaching & Advising

University of Rochester Rochester, New York **2025-Present**

Adjunct Professor/Curriculum Developement

- Physics 100 (Physics and the Natural World) for non-STEM majors, historical development of natural philosophy
- Physics 141 (Honors Mechanics) for incoming physics majors; developed curriculum around Kleppner & Kolenkow.
- Develop curriculum and documentation for all non-honors physics labs; created the department's lab website

University of Rochester Rochester, New York **2023-Present**

Advising/Mentoring

- Advise graduate students S. Kelty (configurational entropy and critical points in the Ising model), B. Bhatt (entropy estimation for fMRI data analysis), and B. Markwick (agent-based foraging simulations).
- Created "My Favorite Spacetimes," an advanced topics seminar for graduate and upper-level undergraduates covering the history of Special Relativity, Minkowski and Rindler spacetimes, and classical papers on GR.
- Founded the Complex Systems and Integrated Information collaboration (2023), coordinating speakers and scheduling.

Teaching & Advising (Continued...)

Thayer School of Engineering, Dartmouth College Hanover, New Hampshire **2019**

Co-Lecturer

- Update curriculum for K. Paulsen's Engineering 105 (finite element methods), deliver lectures during his travel, and hold office hours to help graduate students troubleshoot FORTRAN code.

RIA Global Education Online Course **2019**

Adjunct Professor

- Physics of Music course covering vibrations and waves in different media.

Department of Physics & Astronomy, Dartmouth College Hanover, New Hampshire **2011-2016**

Graduate Student Teaching Assistant

- TA for courses spanning classical mechanics, electromagnetism, introductory astronomy, quantum mechanics (intro and advanced), special and general relativity, cosmology, and quantum field theory.
- Assist teaching Honors Introductory Physics; organize extracurricular mathematics and programming sessions for incoming undergraduates.

Hunter College New York, New York **2010-2011**

Educational Consultant

- Develop and teach a Physics GRE prep course and an intro Python course; organize undergraduate tutoring sessions.

Beginning with Children Foundation New York, New York **2008-2011**

Educational Consultant

- Develop curriculum for after-school tutoring in mathematics and test prep (SSAT, SHSAT, SAT) across multiple NYC charter schools.
- Mentor students on academic habits, oversee and help them with college applications.

Tutor Associates New York, New York **2005-2011**

Head Physics and Mathematics Tutor

- Tutor students in AP Physics and AP Calculus and standardized test prep.
- Develop mock exams, and dynamic spreadsheets to track and analyze student performance over time.

Launch Education *STEM Tutor* New York, New York **2008-2011**

Celia Cruz Bronx High School of Music. *Physics Teacher* New York, New York **2005-2007**

Grace Dodge High School *Mathematics Teacher* New York, New York **2006**

UC Tutors *Tutor* Irvine, California **2004-2005**

SmArT Academy *Physics and Mathematics Teacher* Cerritos, California **2003-2005**

Fullerton Community College *Head Science and Mathematics Tutor.* Cerritos, California **2003-2005**

Cragmont Elementary School *Work Study Teacher's Assistant* Berkeley, California **2001-2003**

Publications (click to link)

Behavioral Speciation in the Neuroevolution of Recurrent Neural Net Foragers driven by Environmental Homogeneity
Damian R. Sowinski
Submitted to *Artificial Life Conference 2026*

The Search for Technosignatures: a Review of Possibilities
Clément Vidal, Benji L. Fields, Damian R. Sowinski, et al
arXiv:2605.21093v1

Causal Leverage Density: A Universal Framework for Semantic Information
Damian R. Sowinski, Silvia Holler, Mike Wong, Gary Segal, David Parkinson, Clément Vidal, James Henderson Cleaves, Pritvik Sinhadc, Anirudh Prabhu, Stuart Bartlett
Artificial Life Conference Proceedings 37. Vol. 2025. No. 1, MIT Press.

Publications (click to link)

Notions of adiabatic drift in the quantized Harper model
Alice C Quillen, Nathan Skerrett, Damian R. Sowinski, Abobakar S. Miakhel
Physical Review A 112 (4), 042226

Exo-Daisy World: Revisiting Gaia Theory through an Informational Architecture Perspective
Damian R Sowinski, Adam Frank, Gourab Ghoshal
The Planetary Science Journal 6.7 (2025): 176.

Information Bounds Production in Replicator Systems
Jordi Piñero, Damian R Sowinski, Gourab Ghoshal, Adam Frank, Artemy Kolchinsky
Nature Communications Physics 9, 120 (2026).

Reflections of Life: Distinguishing Living from Non-living Worlds with Complexity Metrics
Stuart Bartlett, Xianlej San, Siteng Fan, Gary Segal, Damian R Sowinski, Yuk L Yung
(2026). The Astronomical Journal, 172(1), 16

In vivo Magnetic Resonance Imaging of the Interstitial Pressure Gradients using a Pulsatile Poroelastic Computational Model
Matthew McGarry, Damian Sowinski, Likun Tan, Elijah Van Houten, Jaco J.M. Zwanenburg, John Weaver, Keith Paulsen
Interface focus 15 (1), 20240044

Configurational Information Measures, Phase Transitions, and an upper bound on Complexity
Damian Sowinski, Sean Kealty, Adam Frank, Gourab Ghoshal
Under Review, Physical Review E

Semantic Information in a feedback-control Kuramoto Model
Damian Sowinski, Adam Frank, Gourab Ghoshal
Phys. Rev. Research 6, 043188 2024

Semantic Information in a Model of Resource Gathering Agents
Damian Sowinski, Jonathan Carroll-Nellenback, Jordi Pinero, Artemy Kolchinsky, Robert Marwick, Adam Frank, Gourab Ghoshal, Marcelo Gleiser
PRX Life 1 (2), 2024

The Consensus Problem in Polities of Agents with Dissimilar Cognitive Architectures
Damian Sowinski, Jonathan Carroll-Nellenback, Jeremy M DeSilva, Adam Frank, Gourab Ghoshal, Marcelo Gleiser
Entropy 24 (10), 1378

Visual Stream Connectivity Predicts Assessments of Image Quality
Elijah Bowen, Antonio Rodriguez, Damian Sowinski, and Richard Granger
Journal of Vision 22 (11), 4-4

Mapping Heterogenous Anisotropic Tissue Mechanical Properties with Transverse Isotropic Nonlinear Inversion MR Elastography
Matthew McGarry, Elijah Van Houten, Damian Sowinski, Dhrubo Jyoti, Daniel Smith, Diego A Caban-Rivera, Grace McIlvain, Philip Bayly, Curtis L Johnson, John Weaver, Keith Paulsen
Magnetic Resonance in Medicine 87 (3), 1313-1328

Quantifying Stability of Parameter Estimates for In Vivo Nearly Incompressible Transversely-Isotropic Brain MR Elastography
Dhrubo Jyoti, Matthew McGarry, Elijah Van Houten, Damian Sowinski, Philip V Bayly, Curtis L Johnson, Keith Paulsen
Biomedical Physics & Engineering Express 8 (3), 035015

Poroelasticity as a Model of Soft Tissue Structure: Hydraulic Permeability Reconstruction for Magnetic Resonance Elastography In Silico
Damian Sowinski, Matthew McGarry, Scott Gordon-Wylie, Elijah Van Houten, John Weaver, and Keith Paulsen
Frontiers of Physics, Research Topic: Innovative Dev. in Multi-Modality Elastography (2020)

Correlated Noise in Brain Magnetic Resonance Elastography
Ariel J. Hannum, Grace McIlvain, Damian Sowinski, Matthew D. J. McGarry, Curtis L. Johnson
Magnetic Resonance in Medicine (2021) 00: 000– 000.

Publications (click to link)

Towards a Perceptual Distance Metric for Auditory Stimuli
Sarah Oh, Elijah Bowen, Antonio Rodriguez, Damian Sowinski, Eva Childers, Annemarie Brown, Laura Ray, Richard Granger
arXiv:2011.00088

A Heterogenous, Time Harmonic, Nearly Incompressible Transverse Isotropic Finite Element Brain Simulation Platform for MR Elastography
Matthew D J McGarry, Elijah Van Houten, Charlotte Guertler, Ruth J Okamoto, Daniel R Smith, Damian R Sowinski, Curtis L Johnson, Philip Bayly, John Weaver, and Keith D Paulsen
Physics in Medicine & Biology (2020)

Configurational Information Approach to Instantons and False Vacuum Decay in D-dimensional Spacetime
Damian Sowinski , Marcelo Gleiser
Physical Review D, 98, 056026 (2018)

Configurational Entropy as a Lifetime Predictor and Pattern Discriminator for Oscillons
Marcelo Gleiser, Michelle Stephens , and Damian Sowinski
Physical Review D 97, 096007 (2018)

How We Make Sense of the World: Information, Map-Making, and the Scientific Narrative
Marcelo Gleiser, Damian Sowinski
The Map and the Territory: Exploring the Foundations of Science, Thought and Reality, ed. by S. Wuppuluri, N. da Costa, and F. A. Doria, Springer Frontiers Collection (2018)

Information Dynamics at a Phase Transition
Damian Sowinski , Marcelo Gleiser
Journal of Statistical Physics, Volume 167, Issue 5, pages 1221-1232, June 2017

Complexity and Stability for Epistemic Agents: The Foundations and Phenomenology of Configurational Entropy
Damian R. Sowinski
Dartmouth College, Department of Physics and Astronomy, Thesis (2016)

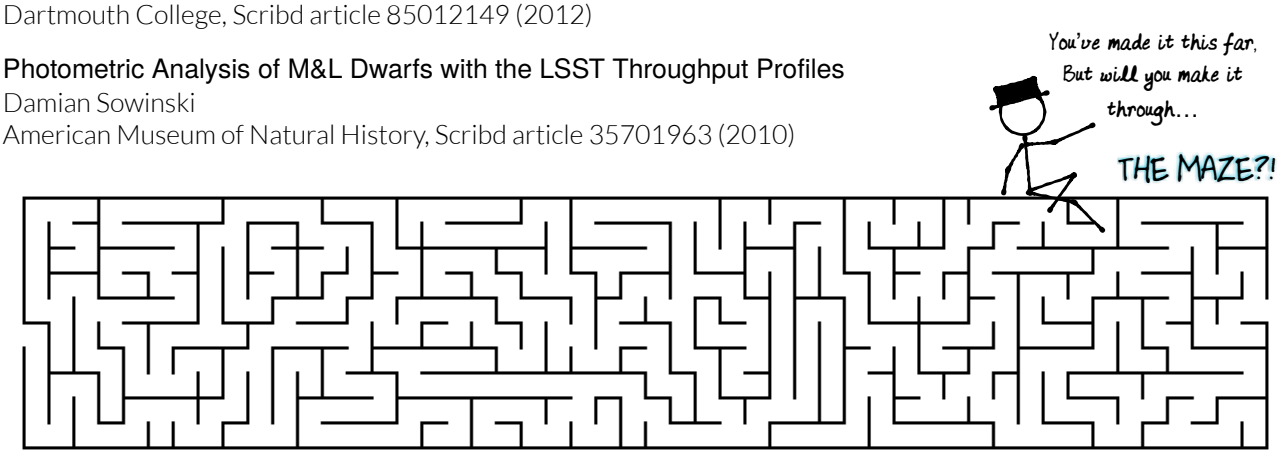
Information-Entropic Signature of the Critical Point
Marcelo Gleiser, Damian Sowinski
Physics Letters B, Volume 747, Pages 125-128, 30 July 2015

Information-Entropic Stability Bound for Compact Objects: Application to Q-Balls and the Chandrasekhar limit of Polytropes
Marcelo Gleiser, Damian Sowinski
Physics Letters B, Volume 727, Issues 1-3, Pages 272-275, 25 November 2013

Cosmic Anisotropy: A History of the Axis of Evil
Damian Sowinski
Dartmouth College, Scribd article 130092552 (2013)

From Monopoles to Textures: A Survey of Topological Defects in Cosmological Quantum Field Theory
Damian Sowinski
Dartmouth College, Scribd article 85012149 (2012)

Photometric Analysis of M&L Dwarfs with the LSST Throughput Profiles
Damian Sowinski
American Museum of Natural History, Scribd article 35701963 (2010)



Science Communications

2025

What is Complexity? Perspectives from Computational Mechanics
University of Rochester, Department of Physics & Astronomy, Seminar

Time Machines and Warp Drives: General Relativity and Sci-fi
McGregor's Tavern, Nerd Night Public Lecture

Top-Down Bottom-Up Causation in Collective Behavior
University of Rochester, Animal Behavior and Cognitive Science, Seminar

Cosmic Strings and the CP Problem
University of Rochester, Astronomy Journal Club

Laplace's Demon and the Drunkard
Bartlett Collaboration on Complexity, Zoom Talk

2024

Time Machines, Wormholes, and Warp Drives
University of Rochester, Astronomy Journal Club

Biospheres and Planets:
Semantic Information Theory & the Gaia Hypothesis
Astrobiology Science Conference

The Meaning in Life: Towards a Theory of Semantic Information
Information Driven States of Matter Workshop,

Information Sharing Polities, Cognitive Architectures, and Consensus
Complex Systems and Information Theory Meeting Seminar

Emergence of Meaning in the Sensor Evolution of Ante-organisms
American Physical Society, March Meeting

Why Can't We Agree?
Biro's Animal Collective Behavior Workshop

My Favorite Spacetimes
University of Rochester, 10 week seminar

Much Ado About Nothing:
Vacuum Phase Transitions & Topological Defects
University of Rochester, Astronomy Journal Club

Consensus in Polities with Heterogeneous
Cognitive Architectures is Hard
Northeast Conference on Complex Systems 2024

2023

Semantic Information in Agent-Based Foraging Models
Complex Systems and Information Theory Seminar

Multi-Timescale Nanoparticle Simulations
Dartmouth College, Nanoparticle Imaging Seminar

Finding Meaning in Finding Food
Northeast Conference on Complex Systems 2023

Naked Singularities in Critical Grav. Collapse
University of Rochester, Astronomy Journal Club

Semantic Information in the Kuramoto Model
Landi's Quantum Information Seminar

2022

Lie Algebras and Conformal Transformations in Phase Transitions
Collaboration with Zajj Daugherty at the City College, NYC

Information Transfer in Ising Models with Temperature Gradients
University of Rochester, Adam Frank's Collaboration Zoom Talk

Simulating Nanoparticle Dynamics
Dartmouth College, Nanoparticle Imaging Seminar

Numerical Techniques for Stochastic DiffeEqs
Dartmouth College, Nanoparticle Imaging Group Seminar

2021

Renormalization Group Flow and the
Acoustoelastic Effect in Gel Phantoms
Dartmouth College, Magnetic Resonance Elastography Zoom Seminar

Langevinference - a novel approach to property inference in MRE
Dartmouth College, Magnetic Resonance Elastography Zoom Seminar

The Acoustoelastic Effect and Hyperelasticity
Dartmouth College, Magnetic Resonance Elastography Zoom Seminar

Information Geometry and Entropic Measures in Data Analysis
Dartmouth College, Psych and Brain Science Department Talk

The Irreducible Representation Expansion of Hyperelasticity
University of Delaware, Anisotropic MRE Collaboration Zoom Talk

Hydraulic Permeability Reconstruction
in Magnetic Resonance Elastography
American Physical Society, March Meeting, Talk

Criticality in the Presence of Physical Gradients
U of Rochester, Adam Frank's Collaboration Zoom Talk

Agent Based Predator-Prey Flocking Models
U of Rochester, Adam Frank's Collaboration Zoom Talk

Coarse Graining in Lotka-Volterra Systems
Sorbonne Universite, Ecological Dynamics Group Z Talk

2020

Diabetic Foot - Biomechanical Aspects of Finite Deformations
Dartmouth College, MRE Zoom Seminar

The Continuum Mechanics of Infinitesimal and Finite Deformations
Dartmouth College, Thayer School of Engineering 2 part Lecture Series

The Foundations of an Effective Field Theory of Tissue
Dartmouth College, Thayer School of Engineering 4 part Lecture Series

Modeling Networks of Interacting Species
Sorbonne Universite, Ecological Dynamics Group Z Talk

A Survey of Biomechanical Models
Dartmouth College, MRE Zoom Seminar

Biomechanics and Arterial Modeling
Dartmouth College, MRE Zoom Seminar

Science Communications (Continued...)

2019

The Weber-Fechner Law and Perception Metrics
Dartmouth College, Psych and Brain Science Department Talk

Information, Quantum Mechanics, and the Eyeball
Dartmouth College, Psych and Brain Science Department Talk

Logit and Probit Functions for Classification in Psychometric Analysis
Dartmouth College, Psych and Brain Science Department Talk

Vector Spaces and Mathematical Methods for Psychophysics
Dartmouth College, Psych and Brain Science 4 part Lecture Series

Regularization Methods in MRE
BioMedical Engineering Society Annual Meeting

A Bayesian Perspective on Optimization
MRE internal annual collaboration meeting

Information Content in Natural Images
Dartmouth College, Psych & Brain Science Department Talk

2018

Do you even Latex?
Kresge Library at Dartmouth College, 4 session Workshop

The ADM Decomposition and Numerical relativity
Dartmouth College, Physics Department Talk

pyCE: A Python Library for Configurational Entropy
Dartmouth College, Physics Department Talk

Steaming Piles of Science: Dr. Sowinski on Information Theory
Interview for a popular science podcast

Oscillons in Vacuum Phase Transitions
American Physical Society, New England Meeting

Finding Boson Stars Numerically
Dartmouth College, Physics Department Talk

An Ode to Oscillons: Criticality in QFT
McGill Northeast Cosmology Workshop, Talk

A Minimal History of Maximal Entropy
Dartmouth College, Postdoc Association Public Lecture

2017

MIB Coordinates and Finite Difference Methods for QFT Simulations
Dartmouth College, 2 Part Lecture Series

Criticality and Information Dynamics in Neuroscience
WISPIC, Neuroscience Department Talk

Ising Models in Neuroscience
WISPIC, Neuroscience Department Talk

2016

From Epistemology to Information
Institute for Cross-Disciplinary Engagement, 3 part Lecture Series

The Epistemic Foundations and Phenomenology of Configurational Entropy
Dartmouth College, Physics Department Lecture

2015

Maximum Entropy Methods in Statistical Physics
Dartmouth College, 2 part Lecture Series

Information Turbulence at Criticality
Carnegie Institute of Science, Poster Session

2014

Probability, Information, and Statistical Physics
Dartmouth College, 4 part Lecture Series

Geometry, Topology, and Physics by Nakahara
Dartmouth College, Reading Course

The Cox Axioms: A Foundational Approach to Bayesian Probability Theory
Dartmouth College, 2 part Lecture Series

2013

Configurational Entropy and the Chandrasekhar Limit
American Physical Society April Meeting, Talk

Cosmic Anisotropy: A History of the Axis of Evil
Dartmouth College, Physics Department Talk

2012

A Multi-Letter Polymerization Model of Chiral Symmetry Breaking
Astrobiology Graduate Student Conference, Poster Session